Experiment 4

Aim: To draw the Data Flow Diagram (DFD) for the Smart Parking Management System using an open-source diagramming tool, representing the flow of data between external entities, system processes, and data stores.

Theory:

Working of the DFD

The **Vehicle Owner** provides credentials to the **User Authentication** process. The system validates the credentials against the user database and generates a JWT session token after successful authentication.

The **Parking Space Status Management** process receives occupancy information from parking sensors and maintains the current status of available, occupied, and reserved spaces. Real-time availability is then provided to the vehicle owner through the application. The SRS specifies that live slot updates should synchronize with client views within two seconds of a status change.

The **Advance Slot Reservation** process receives the user's selected slot, arrival time, and duration. The system temporarily locks the selected slot for 15 minutes and changes its status to **Reserved** after payment confirmation. Redis is used to prevent double-booking during checkout.

The **Navigation Guidance** process uses the active booking and invokes the **Google Maps API** to provide navigation to the selected parking facility.

During entry, the **Gate Entry & Exit Processing** process receives QR scan data from the scanner. The system validates the booking, records the check-in timestamp, and sends a signal to open the gate.

The **Automated Payment Processing** process calculates the parking charges based on the parking duration. The user completes the payment through the payment gateway, after which the transaction status is updated and a digital receipt is generated.

The **Booking History & Receipt Generation** process allows users to view previous parking sessions and generate downloadable PDF receipts.

Finally, the **Facility Management & Dynamic Pricing** process allows the Parking Manager to modify parking rates and block slots for maintenance. Changes are reflected in subsequent parking searches, while confirmed reservations remain unaffected by new pricing rules.

DFD Levels

Level 0 DFD / Context Diagram

At Level 0, the **Smart Parking Management System** is represented as a single process. The external entities interacting with the system are:

- Vehicle Owner
- Parking Manager
- Sensor
- Payment Gateway
- Google Maps API

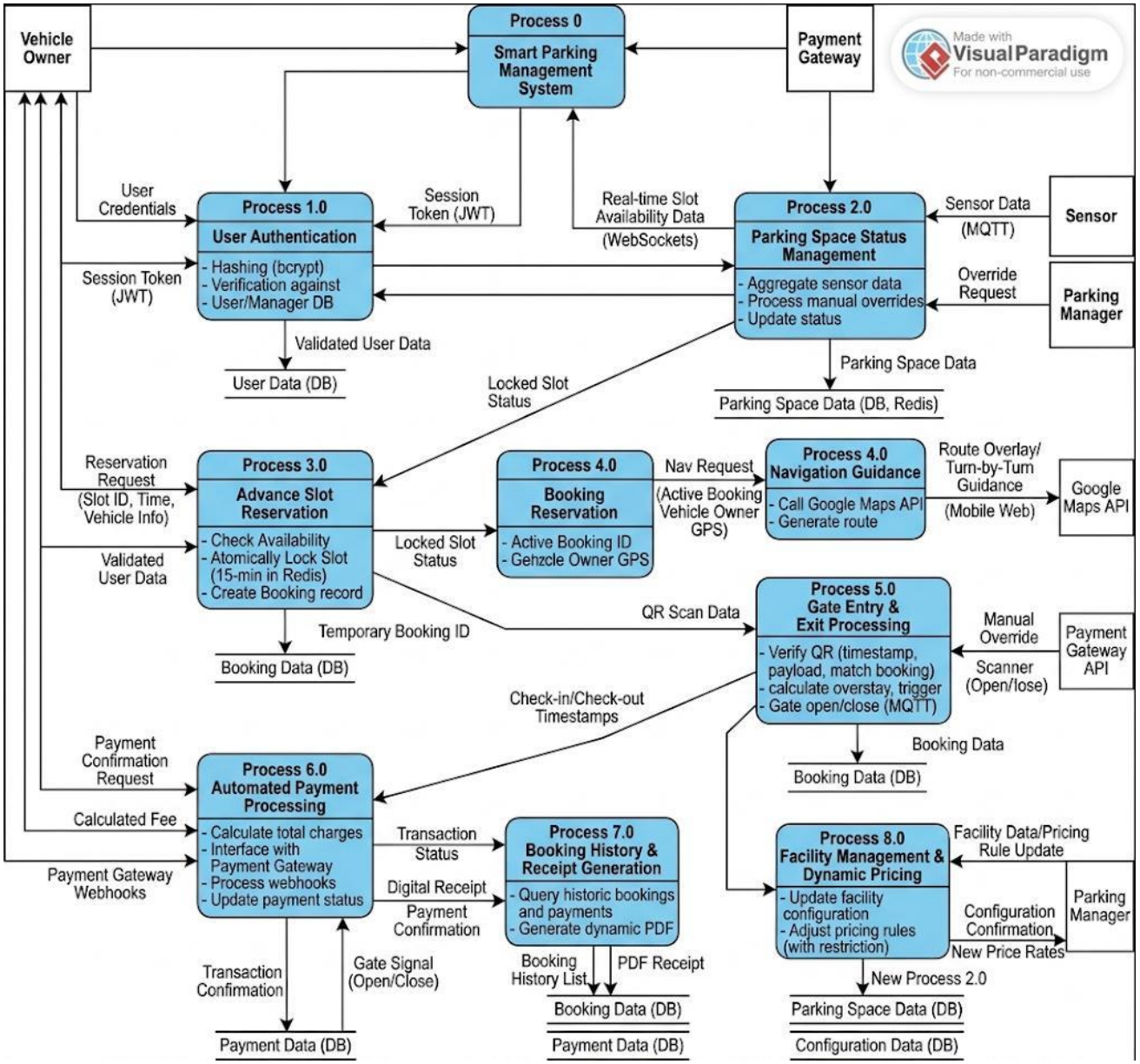
The diagram shows the major input and output data flows between these entities and the complete system.

Level 1 DFD

At Level 1, the main system is decomposed into individual processes such as:

- User Authentication
- Parking Space Status Management
- Advance Slot Reservation
- Booking Reservation
- Navigation Guidance
- Gate Entry & Exit Processing
- Automated Payment Processing
- Booking History & Receipt Generation
- Facility Management & Dynamic Pricing

This provides a more detailed representation of how data moves through the individual functional modules.



Tool Used:

diagrams.net (Draw.io) – an open-source/free diagramming tool used for creating the DFD.

Result:

The Data Flow Diagram for the Smart Parking Management System was successfully designed using an open-source diagramming tool. The DFD represents the flow of data between vehicle owners, parking managers, sensors, external services, system processes, and databases involved in parking reservation, monitoring, navigation, entry/exit, payment, and facility management.